Lesson 07: The Moral Foundations of Ethical Research

English Stage 4 (Psychology) · Unit 2: Research Ethics and Measurement

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 3.1 Prelude to Research Ethics; 3.2 Moral Foundations of Ethical Research.

In this lesson

You learn a framework for thinking about research ethics (four moral principles applied to three groups of people) through three famous cases: the MMR vaccine study, Milgram's obedience study and the Tuskegee syphilis study. You practise the language of balanced ethical argument: concession and the hedged hypothetical past.

Learning goals

After this lesson you should be able to:

- Explain the four moral principles and the three groups affected by research.
- Analyse the MMR, Milgram, Tuskegee and "personal space" cases with the framework.
- Write balanced ethical judgements with concession clauses and hedges.
- Identify the main risks and benefits of your own planned study.

Key language

Term or phrase	Meaning	Example
ethics	The branch of philosophy concerned with morality; also a set of principles and practices that give moral guidance in a field.	There is an ethics of business, medicine and scientific research.
confederate	A researcher who pretends to be someone he or she is not in a study, most often another participant.	The "learner" in Milgram's study was a confederate.
deception	Misleading participants about the purposes and procedures of research, by giving false information or withholding true information.	Some questions are difficult to answer without deception.
debriefing	Informing participants after a study of its purpose, revealing any deception, and minimizing any harm.	Milgram went to great lengths to debrief his participants.
integrity	doing research thoroughly and competently, meeting professional obligations and being truthful	Acting with integrity promotes trust.
justice	treating participants fairly and distributing risks and benefits fairly	Tuskegee is a tragic example of injustice.

Term or phrase	Meaning	Example
autonomy	the right to make your own choices free from coercion	Informed consent respects autonomy.
informed consent	Obtaining and documenting people's agreement to be in a study, having informed them of everything that might reasonably affect their decision.	In neither study did participants give true informed consent.
confidentiality	The researcher's agreement not to reveal participants' personal information except with their permission or as required by law.	Participants trust researchers to keep their promises of confidentiality.
anonymity	names and other identifying information are not collected at all	Ideally, participants can maintain anonymity.
be outweighed by	be less important than	Research is ethical only if its risks are outweighed by its benefits.
it seems likely that	a hedge: the writer believes something but cannot prove it	It seems likely that they would not have agreed to participate.

Rules and explanations

The ideas

The opening case (3.1). In 1998 *The Lancet* published a study claiming a relationship between the measles, mumps and rubella (MMR) vaccine and autism, and many parents stopped vaccinating their children. Follow-up studies consistently failed to find a relationship. The lead researcher also stood to gain financially (he had patented a competing vaccine), used biased methods to select and test participants, and used unapproved, medically unnecessary procedures. In 2010 the article was retracted and he lost his right to practise medicine.

A framework (3.2). A thorough ethical analysis asks how each of four principles (adapted from the APA Ethics Code) applies to each of three groups: **research participants**, the **scientific community** and **society**.

Moral principle	What it requires	Main example
Weighing risks against benefits	Risks must be outweighed by benefits.	Milgram (1963)
Acting responsibly and with integrity	Competent, truthful research that keeps promises and builds trust.	the MMR study
Seeking justice	Fair treatment and compensation; risks and benefits fairly distributed.	Tuskegee (1932-1972)
Respecting rights and dignity	Autonomy (informed consent) and privacy (confidentiality, anonymity).	Tuskegee, Milgram

Weighing risks against benefits. Participants risk useless or harmful treatments, physical or psychological harm and loss of privacy; they may gain a helpful treatment, knowledge about psychology, satisfaction, or money or course credit. A poorly designed study wastes science's time and money; misunderstood or misapplied results can harm society. Weighing is hard because the risks often fall on participants while the benefits go to science or society. In **Milgram's obedience study**, participants believed they were giving stronger and stronger shocks to another participant, who protested, screamed and fell silent; he was a **confederate**, and the screams were a recording. Most participants kept giving the shocks right through the protests. The result is considered one of the most important in psychology, but Milgram himself reported extreme tension (14 of 40 participants showed nervous laughter; three had uncontrollable seizures of laughter). To his credit, he debriefed them carefully.

Acting with integrity. Participants, science and society must be able to trust researchers. **Deception** creates a conflict: some questions, like Milgram's, are difficult or impossible to answer without it. After the MMR study, researchers wasted resources on follow-ups and children were put at risk.

Seeking justice. If a new psychotherapy works, it is fair to offer it to the control group afterwards. At the level of society, some groups (institutionalized or disabled people, racial and ethnic minorities) have carried more than their fair share of research risks. In the **Tuskegee study** (US Public Health Service, 1932-1972), poor African American men were told they were being treated for "bad blood" but were never treated for syphilis, even after penicillin became standard in the 1940s. In 1997 President Clinton apologized: "They were betrayed." (The book's CDC link to the apology no longer works; the full text is in the White House archive: see Watch online.)

Respecting rights and dignity. Autonomy requires **informed consent**, which neither the Tuskegee men nor Milgram's participants truly gave. Privacy requires **confidentiality** and, ideally, **anonymity**.

Unavoidable conflict. Almost no research is risk-free, and some conflicts divide competent researchers. In a "personal space" study in a public men's room (Middlemist, Knowles, & Matter, 1976), researchers secretly observed whether men took longer to begin urinating with a confederate nearby. A critic called it an unjustified assault on human dignity (Koocher, 1977); the researchers had weighed the issues. The responsible response: think the issues through, minimize and weigh the risks, explain your decisions, seek feedback and take responsibility.

Language focus

1. Concession. The writer accepts one point (in the *although/while* clause) and then makes the more important one (in the main clause).

- "**Although** this is considered one of the most important results in psychology, it came at the cost of producing severe psychological stress in the research participants."
- "**While** some critics found this to be an unjustified assault on human dignity, the researchers had carefully considered the ethical conflicts."

2. The hedged hypothetical past. "**Had they been told** this basic fact about the study, **it seems likely that** they **would not have agreed** to participate." Form: *Had* + subject + past participle, ...

would (not) have + past participle (= "If they had been told ..."). The hedge admits that no one can prove what people would have done.

> **Note for Arabic speakers** - Do not use *although* and *but* together ("Although they agreed, but ..."). Use one: "Although they agreed, they were not informed." - For unreal past situations, use the past perfect and *would have* + past participle: "If they had been told, they would not have agreed", not "If they were told, they would not agree".

Practice

1. Reading questions (sections 3.1 and 3.2)

1. Apart from the failed follow-up studies, what three problems were found in the 1998 MMR research, and what happened in 2010?
2. What must a thorough consideration of the ethics of a research project take into account?
3. Why is it "not necessarily easy" to weigh risks against benefits?
4. In Milgram's study, who was the other "participant", and where did the screams come from?
5. Why can acting with integrity conflict with advancing scientific knowledge?
6. Who were the Tuskegee participants, what were they told, and what was withheld from them?
7. What is the difference between confidentiality and anonymity?
8. In the "personal space" study, what did the critic say, and what evidence did the researchers give?

2. Applying the framework (sections 3.1 and 3.2 have no exercises; the chapter's exercises are in Lesson 09)

1. Place each case in the framework (principle and group). More than one answer can be right; give a reason. a. Milgram's participants sweated, trembled and stuttered. b. Researchers spent resources on unnecessary MMR follow-up studies. c. Parents chose not to vaccinate their children. d. An effective therapy is offered to the control group after the study. e. Poor African American men were left untreated for decades. f. The Tuskegee men were never told they had syphilis.
2. The book asks about Milgram's study: "Was It Worth It?" Answer in about 100 words, using at least two principles.

3. Language A. Join each pair with *although* or *while*, putting the accepted point in the subordinate clause.

1. The Tuskegee participants were given some free medical care. They were not treated for their syphilis.
2. Most of Milgram's participants were glad to have taken part. Many experienced severe stress.

B. Correct: "Although Milgram debriefed his participants, but they did not give true informed consent."

C. Rewrite with *Had* and *it seems likely that*.

1. If Milgram's participants had been told that they might be "reduced to a twitching, stuttering wreck," many of them would probably not have agreed to participate.

2. If the lead MMR researcher had reported honestly, other researchers would probably not have wasted resources on follow-up research.

Speak with your tutor

- "Let me explain the four-principles, three-groups framework using the MMR case. Correct my explanation and my English."
- "Argue that Milgram's study was worth it. I will reply with sentences beginning with *Although* or *While*."
- "In PSYC 110 Lecture 15 (Morality), Professor Bloom says that 'Milgram's experiment is why we have human-subjects committees'. Help me compare his criticism with the book's view of Milgram's debriefing."
- "In PSYC 110 Lecture 17, Professor Bloom says that keeping people from sleeping would cause 'tremendous pain and anguish'. What does this mean for the risks of my sleep and memory study?"

Writing task

Begin the ethics section of your proposal (sleep and memory in university students, or your own topic): write a **risk-benefit analysis** of the study you outlined in Lesson 06 (200-300 words).

- One sentence summarising the study (default: 20 students learn 40 word pairs; half, chosen at random, have a nap opportunity of up to 90 minutes, and the other half rest quietly awake for the same time; then all try to recall the pairs).
- Risks and benefits for participants (for example, tiredness or boredom in the wake group, discomfort at sleeping in a laboratory, stress during the memory test; course credit, learning about psychology).
- At least one risk or benefit for science and one for society.
- One point for each other principle: integrity (honesty, any deception), justice (fair compensation, who carries the risks), rights and dignity (consent, confidentiality or anonymity).
- A conclusion: do the benefits outweigh the risks, and how could you reduce the risks?
- Language: at least two concession sentences and one hedge.

If you work on your own topic, analyse your own study in the same way.

Self-editing checklist:

- Did I cover participants, science and society?
- Did I mention all four principles?
- Are the risks specific to my study?
- Does each *although/while* clause hold the point I accept?
- Did I avoid *although … but*?
- Did I hedge claims I cannot prove?
- Does my conclusion say how I would reduce the risks?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- President Clinton's 1997 apology for the Tuskegee study, full text (White House archive): https://clintonwhitehouse4.archives.gov/textonly/New/Remarks/Fri/19970516-898.html
- Healthcare Triage, "Vaccines Don't Cause Autism" (the video in section 3.1): https://www.youtube.com/watch?v=065l1YAVaYc
- *The New York Times* (2010) on the doctor who linked the vaccine to autism (may need a subscription): https://www.nytimes.com/2010/05/25/health/policy/25autism.html

Answer key

1. Reading questions

1. He stood to gain financially (he had patented a competing measles vaccine); he used biased methods to select and test participants; he used unapproved and medically unnecessary procedures. In 2010 *The Lancet* retracted the article and his right to practise medicine was revoked (3.1).
2. How each of the four moral principles applies to each of the three groups: participants, the scientific community and society.
3. They may not be directly comparable: the risks are often mainly to participants, the benefits mainly to science or society.
4. A confederate (a helper pretending to be a participant); the screams were an audio recording activated when the participant flipped the switch.
5. Some questions, such as Milgram's, are difficult or impossible to answer without deceiving participants.
6. Poor African American men near Tuskegee, Alabama, told they were being treated for "bad blood". They were not told they had syphilis, were not treated for it (even after penicillin became standard in the 1940s) and were not given an opportunity to leave.
7. Confidentiality: personal information is collected but not disclosed without consent or legal authorization. Anonymity: names and identifying information are not collected at all.
8. The critic (Koocher, 1977) called it an unjustified assault on human dignity. The researchers had interviewed preliminary participants, and none was bothered at having been observed.

2. Applying the framework

1. (Other reasoned answers are possible.) a. Weighing risks; participants. b. Integrity (trust was violated) or weighing risks; scientific community. c. Integrity or weighing risks (results misapplied); society. d. Justice; participants. e. Justice; society (a group carrying more than its fair share of risk), and participants. f. Rights and dignity (no true informed consent); participants.
2. (example answer) Although Milgram's findings help us understand events such as the Holocaust, I do not think they were worth the harm. The risks fell on the participants, who suffered extreme stress, while the benefits went to science and society. The participants also did not give true informed consent. His careful debriefing reduced the harm but could not undo it. (Arguing "yes" is also possible if you give more weight to the knowledge gained.)

3. Language A. 1. Although they were given some free medical care, they were not treated for their syphilis (the book's sentence). 2. Although (or While) most of Milgram's participants were glad to have taken part, many experienced severe stress. B. Although Milgram debriefed his participants, they did not give true informed consent. C. 1. Had participants in Milgram's study been told that they might be "reduced to a twitching, stuttering wreck," it seems likely that many of them would not have agreed to participate (the book's sentence). 2. Had the lead MMR researcher reported honestly, it seems likely that other researchers would not have wasted resources on follow-up research.

My notes